

DATA ANALYST ASSESSMENT — NYC YELLOW TAXI ANALYSIS	
Question 1 — Dataset Selection & Business Context	
Dataset name	NYC Yellow Taxi Trip Data — 200,000-row reproducible sample
Primary source	NYC Taxi & Limousine Commission (TLC) Trip Record Data
Primary source URL	https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page
Analysis/sample URL	https://datatweets.com/datasets/nyc-taxi/yellow_tripdata_sample.csv
Rows in analysis sample	200000
Columns in analysis sample	10
Format	CSV
Description	200,000 real NYC yellow-taxi trips drawn from TLC public trip records; 10 operational/fare fields.
Why selected	Large enough for meaningful processing, includes time, location, distance, payment and monetary fields, and supports operational/revenue questions.
Business opportunities	Payment-channel mix, fare concentration, high-value trip monitoring, demand/operations planning, data-quality controls.

RAW DATA NOTE

The full 200,000-row source file is intentionally not fabricated or reconstructed here. Use the notebook/script in this pack to download the exact public sample into this worksheet. A 5-row schema preview is provided below.

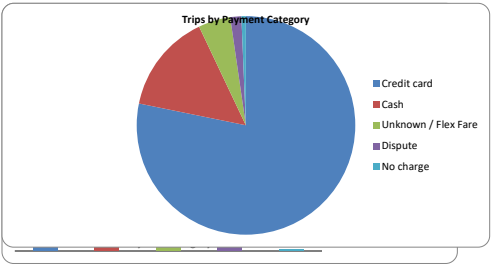
tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	PULocationID	DOLocationID	payment_type	fare_amount	tip_amount	total_amount
2024-01-01 00:00:00	2024-01-01 00:20:00	2	17.14	138	132	1	70	14	86
2024-01-01 00:10:00	2024-01-01 00:20:00	1	2.49	161	237	1	13.5	3	18
2024-01-01 00:20:00	2024-01-01 00:30:00	2	1.84	237	236	2	10	0	13.5
2024-01-01 00:30:00	2024-01-01 00:40:00	1	3.6	236	162	1	23.3	4	29.3
2024-01-01 00:40:00	2024-01-01 00:45:00	1	0.04	162	161	1	3.7	1	6.2

A. Business problem / opportunity	Understand how trip value and payment behavior are distributed so an operations/technology team can prioritize payment reliability, high-value trip workflows, and monitoring.
B1	What share of trips use each payment category?
B2	What is the average trip value and average trip distance?
B3	How large is the high-fare trip segment (total_amount > \$50)?
B4	Where are the largest data-quality gaps that could distort operational KPIs?
B5	Which business actions should be prioritized based on the observed mix and concentration?
C1 Hypothesis	Credit-card payments are the dominant payment channel.
C2 Hypothesis	A relatively small high-fare segment contributes disproportionate operational value and deserves separate monitoring.

Change	Why necessary?	What happens if not done?
Parse pickup/dropoff timestamps	Time-based analysis requires true datetime types.	Hourly/day analysis becomes error-prone and sorting may be lexical rather than chronological.
Handle missing passenger_count	Passenger count has 9,511 missing values in the sample.	Passenger-based KPIs could be biased if missing values are silently treated as zero.
Validate numeric ranges / outliers	Trip distance, fares and totals can contain unusual values.	Extreme records can distort averages and management conclusions.
Map payment codes to labels	Numeric codes are hard for management to interpret.	Dashboard categories become opaque and error-prone.
Create trip_duration_min, avg_value and high_fare_flag	Makes operational and value segmentation explicit.	Repeated ad-hoc calculations increase inconsistency.
Remove exact duplicates if present	Duplicate records can inflate trips and revenue.	Volume and revenue KPIs can be overstated.

Metric	Value	Definition / calculation
Rows	200000	Sample row count
Columns	10	Sample column count
Total revenue	5343779.62	Sum of total_amount
Average trip value	26.7188981	total revenue / trips
Average trip distance (mi)	3.582	Mean trip_distance
High-fare trips	12675	total_amount > \$50
High-fare share	0.063375	high-fare trips / total trips
Credit-card share	0.78189	payment_type = 1
Cash share	0.14761	payment_type = 2
Missing passenger_count	9511	Null/missing passenger_count

Payment category	Trips	Share
Credit card	156378	78.2%
Cash	29522	14.8%
Unknown / Flex	9511	4.8%
Fare		
Dispute	3235	1.6%
No charge	1354	0.7%



Insight	Evidence	Why relevant to management	Business impact	Recommendation
Credit card is the dominant payment channel.	156,378 of 200,000 trips = 78.2%.	Payment technology reliability directly affects most trips.	Failures in the dominant channel can affect a large share of transactions.	Prioritize credit-card transaction monitoring, uptime and failure-rate alerts.
Cash remains a meaningful secondary channel.	29,522 trips = 14.8%.	Cash is large enough to matter operationally.	Cash workflows need reconciliation and exception controls.	Track cash share and reconciliation exceptions by operating period.
High-fare trips are a minority segment.	12,675 trips = 6.3% have total_amount > \$50.	A smaller segment can require different monitoring and service rules.	Targeted controls can focus effort without burdening every trip.	Create a high-value-trip monitoring view and investigate extreme fares.
Average trip value is about \$26.72.	\$26.72 average total_amount across 200,000 trips.	Provides a simple management benchmark for revenue per trip.	Changes in mix or pricing can move revenue per trip even if trip volume is stable.	Track average trip value weekly/monthly and segment by payment and distance.
Average trip distance is 3.582 miles.	Mean trip_distance = 3.582 miles.	Distance is a useful operational proxy for trip complexity and expected value.	Routing, service time and fare mix can vary materially with trip length.	Monitor value and operational KPIs by distance band in the dashboard.

Unexpected result	The payment mix is highly concentrated: one payment category (credit card) accounts for about four-fifths of the sample.
Initial expectation	Payment methods might be more evenly distributed across categories.
What data showed	Credit card = 78.2%; cash = 14.8%; remaining categories make up the balance.
Why this may happen	The dataset reflects electronic taxi transactions and the sampled population may have a payment mix influenced by rider behavior, location and trip context.
Additional analysis performed	Compared counts and shares by payment code; separately flagged high-fare trips to avoid relying on volume alone.
Reasonable conclusion	Payment technology performance should be treated as a core operational KPI, while cash remains important enough to monitor.
Caution	The public trip data does not identify customer-level causes, so the exact reason for the mix cannot be established from this dataset alone.

Issue / risk	Potential effect	Handling / accounting for it
Missing passenger_count (9,511 rows in sample)	Passenger-based averages can be biased.	Do not impute missing passenger counts as zero; exclude missing values from passenger-specific metrics and report completeness.
Extreme / unusual fare or distance records	Means and totals can be distorted by outliers.	Use validation rules, percentile views and a high-fare flag; investigate extreme records separately.
Payment-code semantics	Incorrect mapping can misclassify transactions.	Use TLC's data dictionary and document the code-to-label mapping.
Public-source accuracy limitation	TLC states it does not guarantee the accuracy of submitted trip data.	Treat findings as operational signals, not audited financial statements.
Sample limitation	A 200k-row reproducible sample may not represent the entire NYC taxi population.	Use the sample for assessment analysis; rerun on the full monthly file before production decisions.

Two limitations

1	The analysis sample is not the complete TLC monthly population.
2	The public data lacks customer-level context, profitability/costs and causal explanations.

Conclusion that cannot safely be made

We cannot conclude that a particular payment method causes higher revenue or profitability without controlling for trip distance, location, time and other confounders.

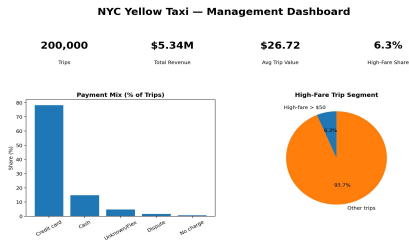
Priority	Recommendation	Supporting insight	Owner	Potential outcome	Success metric
1 - High	Build payment reliability monitoring focused on credit-card transactions.	Credit card is 78.2% of trips.	Product/Engineering + Operations	Lower payment failures and fewer transaction exceptions.	Payment success rate, failure rate, incident minutes.
2 - High	Create a high-value trip exception workflow for trips > \$50.	12,675 trips (6.3%) exceed \$50.	Operations + Data	Faster identification of anomalous/high-value transactions.	High-fare exception rate, confirmed anomaly rate, resolution time.
3 - Medium	Establish a recurring KPI pack for trip value, distance and payment mix.	Average trip value is \$26.72; average distance is 3.582 mi.	BI/Analytics + Management	Earlier detection of mix shifts and operational changes.	Weekly KPI variance, dashboard adoption, alert response time.



Dashboard objective
Key KPIs
Major visualizations
Recommended filters
Looker Studio status

QUESTION 8 — MANAGEMENT DASHBOARD

Give management a compact view of trip volume, revenue, payment mix and high-value trip monitoring.
Trips | Total Revenue | Average Trip Value | High-Fare Share
Payment Mix (%) and High-Fare vs Other Trips
Date | Payment Category | Pickup Zone | Distance Band
Dashboard-ready visual included below. A shareable Looker Studio link requires uploading the processed dataset to Google Sheets/Looker Studio.



AI tools used	Microsoft Copilot and online documentation/resources.
What used for	Understanding the structure of the dataset Getting guidance for Python and Pandas functions Debugging and improving parts of the data-processing code Brainstorming possible business questions and visualization ideas Improving the clarity and presentation of the findings
Example where AI helped	AI helped me understand how to clean the dataset, handle missing values, create calculated fields, and summarize the data for business analysis.
Verification / correction example	I reviewed the generated suggestions and code before using them. I verified the calculations and dataset outputs myself and modified the analysis according to the requirements of the assessment and the actual results obtained from the data.
Responsibility note	AI was used as a learning and productivity tool, while the final dataset analysis, interpretation of results, and submission structure were reviewed and prepared according to the assessment requirements.